PUMP_UNIT_01

Pump Training Manual

Operator triage and maintenance handoff for the simulated pump.

Scope

Use this reference to demonstrate alert review on the simulated telemetry feed. For physical equipment, the site's procedures and the manufacturer's equipment manual govern shutdown, isolation, inspection, repair, and restart.

Troubleshooting sections

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Energy control reference

OSHA 29 CFR 1910.147, control of hazardous energy

General triage and safe handoff

Use this checklist when a model alert has unclear symptoms or several possible causes.

Checklist

1. Confirm the pump ID, alert time, and signal source. Check the latest readings against the operating baseline.

2. Verify signal quality across vibration, temperature, pressure, and motor current. Record the readings and any recent operating changes.

3. Escalate a sudden change or repeated alert through the site's operations procedure. Follow the site's shutdown decision process.

4. Before physical servicing, have an authorized person follow site energy-control procedures and verify isolation, including stored pressure. Follow the manufacturer's procedures for repair and restart.

Escalation record

Record the alert time, verified channels, operating conditions, and the person receiving the handoff.

Bearing wear

Look for a persistent vibration change and higher motor current at comparable operating load.

Checklist

1. Compare vibration and motor current against recent readings at the same load. Check whether the change persists.

2. Review the lubrication history, mounting records, and recent bearing work. Escalate recurring abnormal vibration to qualified maintenance.

3. After verified site isolation, have qualified maintenance inspect bearing condition, mounting, alignment, and lubrication using the manufacturer's instructions.

4. Record the repair decision and compare the readings with the approved baseline during a supervised return to service.

Escalation record

Record the vibration and motor-current trend, load, maintenance history, and findings.

Overheating

Look for a temperature rise and changes in motor current or operating load.

Checklist

1. Cross-check temperature with recent load and motor current. Verify the quality of the temperature signal.

2. Review cooling and ventilation indications available to the operator. Escalate a persistent temperature rise through site procedures.

3. After verified site isolation, have qualified maintenance inspect cooling restrictions, electrical load history, and motor condition using the manufacturer's instructions.

4. Record the repair decision and verify a stable temperature using the approved return-to-service checks.

Escalation record

Record the temperature and current trend, load, cooling indications, and the maintenance findings.

Cavitation

Look for reduced pressure and increased vibration at the current operating point.

Checklist

1. Check whether the pressure drop and the increase in vibration occurred together. Record the current operating point.

2. Review suction pressure, level, flow, and the available suction-line indications. Escalate a persistent change to operations.

3. After verified isolation and depressurization, have qualified maintenance inspect for inlet restriction, air ingress, or impeller damage using the manufacturer's instructions.

4. Record the suspected cause and verify the operating point with site-approved checks before return to service.

Escalation record

Record the pressure and vibration trend, flow and level readings, and any changes to the suction conditions.